MARTA KOWAL-BRANDT

QA Automation Engineer | CI/CD & Kubernetes Test Infrastructure

PROFESSIONAL SUMMARY

Owns GitLab CI and the design, deployment, and lifecycle automation of Kubernetes-based ephemeral test environments for three teams, using Helm, PostgreSQL, seeded data, and end-to-end testing. QA and product teams use these environments to review features before merge. Works with namespace-level access on a shared Kubernetes cluster operated by the platform team. Delivered measurable improvements to pipeline speed and CI costs, and built observability dashboards that helped teams reduce flaky tests. Developing further in cloud infrastructure and platform engineering through hands-on Kubernetes, GitOps, Terraform, and observability projects.

TECHNICAL SKILLS

CI/CD: GitLab CI, Jenkins, pipeline optimization, dependency caching, parallel test execution

Containers and orchestration: Docker, Kubernetes, k3s, Helm

Infrastructure as Code and automation: Terraform (personal-project use), Ansible (basic), Bash

GitOps: Argo CD

Monitoring and observability: Prometheus, Grafana, Loki

Networking and TLS: Traefik ingress; cert-manager; TLS certificate management

Programming and testing: Python, Java, Selenium, pytest, pytest-xdist

Databases and integrations: PostgreSQL, GitLab API

Cloud: AWS — EC2 and S3 fundamentals through learning; no professional AWS production experience. Hetzner VPS provisioned with Terraform for a personal project.

PROFESSIONAL EXPERIENCE

QA Automation Engineer (Test Infrastructure) | Odrava Health Software *2021–Present*

- Own GitLab CI pipelines and the delivery and lifecycle automation of ephemeral test environments used by three teams.
- Built merge-request environments comprising an application, PostgreSQL, and seeded data, deployed via Helm to a shared Kubernetes cluster from GitLab CI.
- Configured environments to be automatically removed after merge or 48 hours of inactivity; QA and product teams use them to review features before merge, and end-to-end tests run against them.
- Work with namespace-level access on the shared cluster; the platform team operates the Kubernetes cluster.
- Profiled pipelines across three teams, parallelized test jobs, added dependency caching, and separated end-to-end tests into their own stage. Reduced average pipeline time from 38 to 14 minutes and monthly CI runner costs by about 30%.
- Built a Python and PostgreSQL process to collect CI test results and created Grafana dashboards to highlight flaky tests by team. After teams addressed leading offenders, the flaky-test rate fell from 7% to below 1%.
- Documented the internal CI operating model in a guide used for onboarding.
- Mentored two QA interns in 2023.

QA Engineer | Brightmill Software House *2019–2021*

- Tested web and mobile applications for multiple clients.
- Introduced the first automated regression suite for the largest project, using Java and Selenium, and configured Jenkins to run it nightly.

SELECTED PROJECTS

Homelab k3s Cluster with GitOps | Personal Project | *2024–2025*

- Provisioned a small Hetzner VPS with Terraform and operated applications through GitOps on a three-node Raspberry Pi 5 k3s cluster; configured Argo CD, Traefik ingress, cert-manager for TLS certificate management, and Prometheus, Grafana, and Loki monitoring.
- Technologies: k3s, Argo CD, Prometheus, Grafana, Loki, Terraform, Ansible (basic).

Merge-Request Ephemeral Environments | Odrava Health | *2023–2024*

- Created short-lived merge-request environments for feature reviews and end-to-end testing, with automated cleanup after merge or 48 hours of inactivity.
- Technologies: GitLab CI, Helm, Kubernetes, Docker, Bash, PostgreSQL.

Flaky Test Dashboard | Odrava Health | *2023*

- Collected CI test results in PostgreSQL and built Grafana dashboards showing flaky tests by team; the flaky-test rate fell from 7% to below 1% after teams addressed leading offenders.
- Technologies: Python, PostgreSQL, Grafana, GitLab API.

CI Pipeline Speed-up | Odrava Health | *2022*

- Reduced average pipeline time from 38 to 14 minutes and monthly CI runner costs by about 30% across three teams.
- Technologies: GitLab CI, Docker, pytest-xdist.

CERTIFICATIONS

- ISTQB Foundation Level, 2019

PROFESSIONAL DEVELOPMENT

- Certified Kubernetes Administrator (CKA) — exam scheduled for January 2027; not yet certified
- HashiCorp Terraform Associate — practice tests completed; exam not taken.